

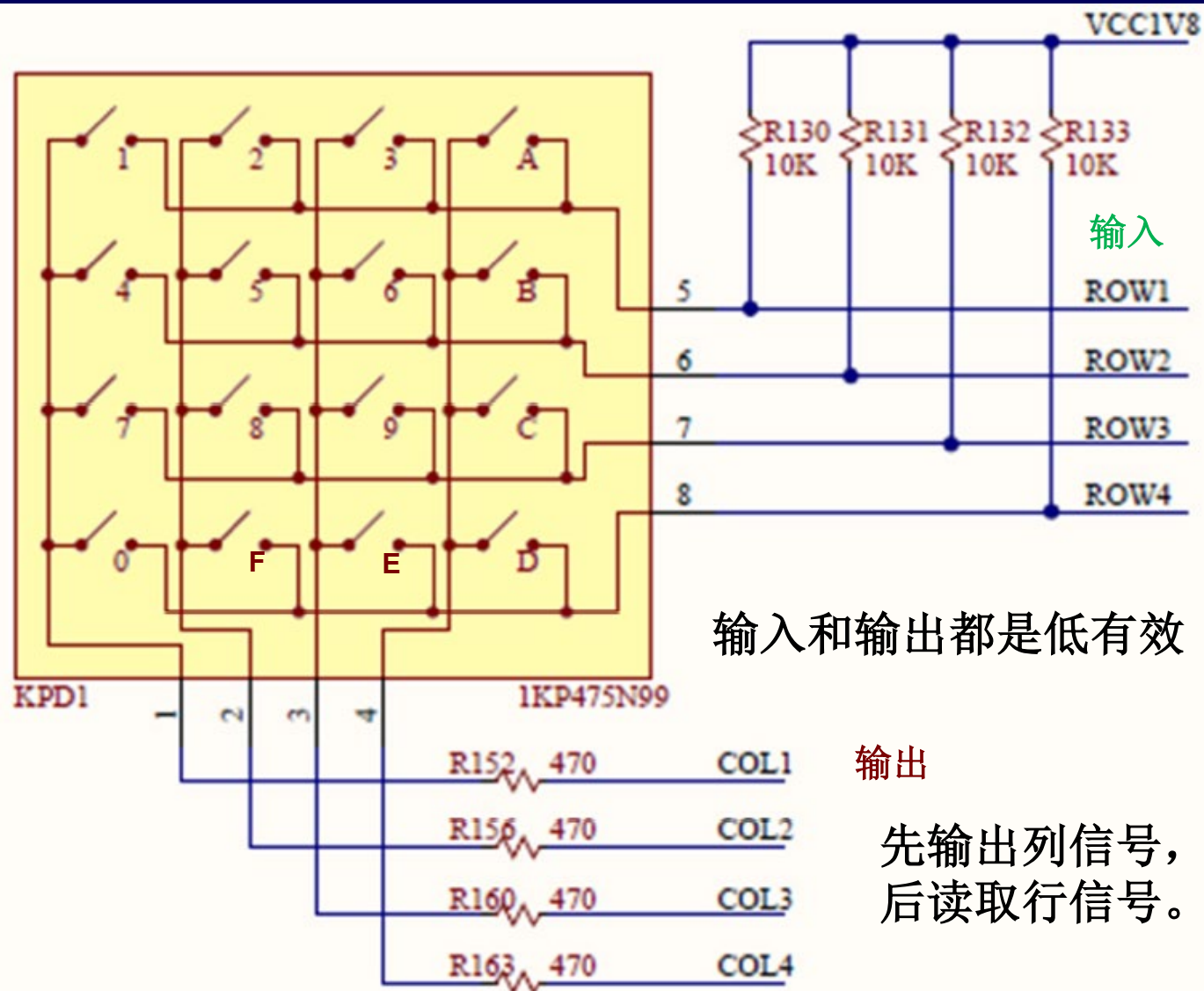
实验任务

4×4小键盘控制器的设计和实现。

实验要求如下：

1. 4×4小键盘的扫描输入。
2. 时钟分频器的设计。
3. 用Led灯显示4×4小键盘的输入键值。

4 × 4 Keypad

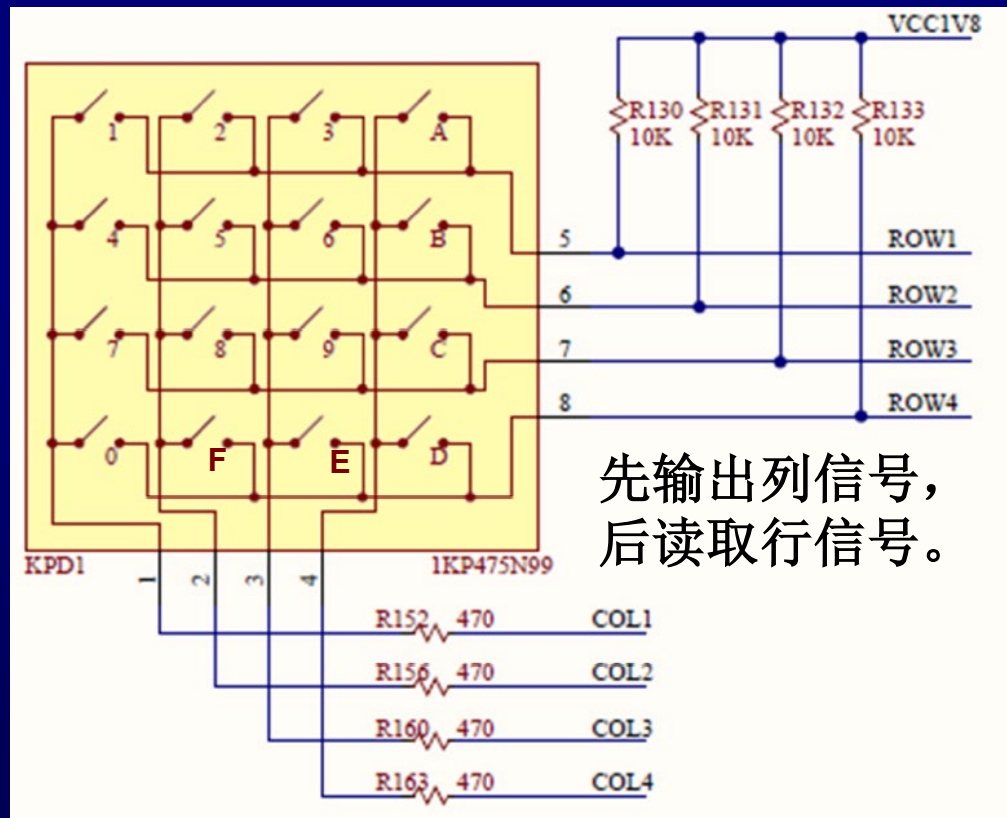


4 X 4 Keypad

```

case( column )
  4'b1110 :
    case( Row )
      4'b1110 : begin KeypadData <= 8'h_1; end
      4'b1101 : begin KeypadData <= 8'h_4; end
      4'b1011 : begin KeypadData <= 8'h_7; end
      4'b0111 : begin KeypadData <= 8'h_0; end
    endcase
endcase

```



时钟分频器

```
module ClockDivider(  
    input      Reset ,  
    input      Clock ,  
    output reg DividedClock    );  
    parameter COUNTER = 32'hC350; // 50 000  
    reg [32:0] clock_count;  
    always @(posedge Clock)  
        if (Reset == 1'b1)  
            clock_count <= {32{1'b0}};  
        else  
            begin  
                if ( clock_count == COUNTER-1 ) // CLK为100M时,  
dividedClk的周期 1ms , 高低各 0.5ms  
                    begin  
                        DividedClock <= ~DividedClock;  
                        clock_count <= {32{1'b0}};  
                    end  
                else  
                    clock_count <= clock_count + 1;  
                end  
            end  
endmodule
```

顶层模块

```
module Keypad_Anvyl(  
    input          CLK ,  
    input  [3:0] KYPD_ROW ,  
    output [3:0] KYPD_COL ,  
    output [7:0] LED  
);  
    wire Clock_1KHz ;  
    ClockDivider clock_divider(  
        .Reset      ( 1'b0 ) ,  
        .Clock       ( CLK ) ,  
        .DividedClock( Clock_1KHz )  
    );  
    Keypad keypad(  
        .Reset_N      ( 1'b1 ) ,  
        .Clock         ( Clock_1KHz ) ,  
        .Row           ( KYPD_ROW ) ,  
        .Col           ( KYPD_COL ) ,  
        .KeypadData    ( LED )  
    );  
endmodule
```

约束文件

```
NET CLK          LOC=D11 ;
## LEDs
NET "LED[0]"     LOC=W3  ;
NET "LED[1]"     LOC=Y4  ;
NET "LED[2]"     LOC=Y1  ;
NET "LED[3]"     LOC=Y3  ;
NET "LED[4]"     LOC=AB4  ;
NET "LED[5]"     LOC=W1  ;
NET "LED[6]"     LOC=AB3  ;
NET "LED[7]"     LOC=AA4  ;
## Keypad
NET "KYPD_COL[0]" LOC = H8 ;
NET "KYPD_COL[1]" LOC = J7 ;
NET "KYPD_COL[2]" LOC = K8 ;
NET "KYPD_COL[3]" LOC = K7 ;
NET "KYPD_ROW[0]" LOC = E4 ;
NET "KYPD_ROW[1]" LOC = F3 ;
NET "KYPD_ROW[2]" LOC = G8 ;
NET "KYPD_ROW[3]" LOC = G7 ;
```

实际测试

